Ex.No.-4.1-Cut section of solids

AIM:

To create the solid model from given drawing and section the solid model

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10 3.16 GB of RAM (integrated graphics recommend 1 GB or more)
3. 2.5 Mbps or faster download; 500 Kbps or faster upload

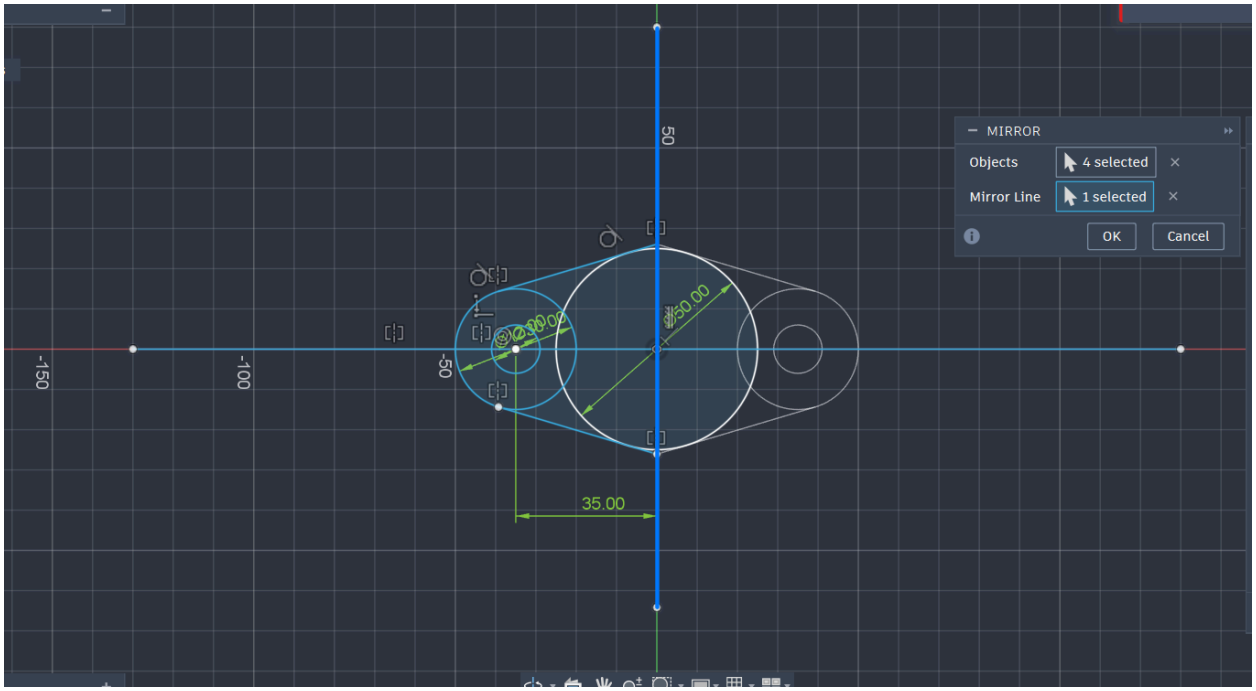
PROCEDURE:

STEP-1:
Open new file and then select sketch icon.
Select the top view for sketching.

STEP-2:
Draw a circle of diameter 50mm at origin.
On the left, draw two circles of diameter 12mm and 30mm, having same centre.
The distance between the centres of the circles is 35mm.

STEP-3:
Create a line such that it is tangent to both circles.

Use the mirror tool to copy the image about the x and y axes.



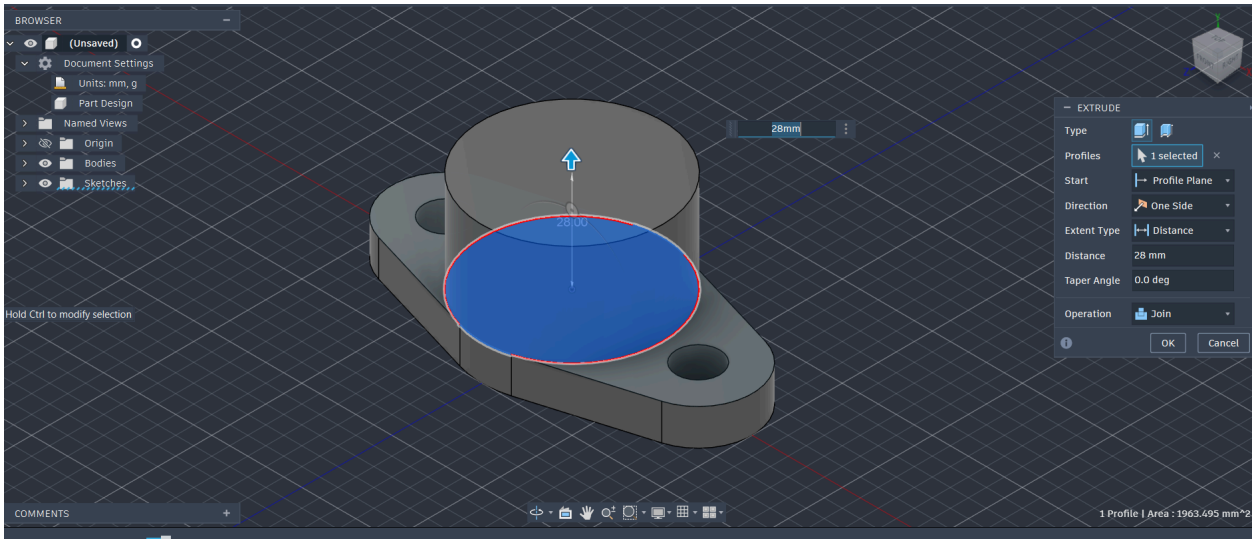
Trim the excess.
After finishing the sketch use finish sketch option to finalize the sketch.

STEP-4:
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of 10mm.
Press ENTER to accept the value.

STEP-5:
Start the sketch command.
Select Sketch > Create Sketch.
Select the Sketch Plane.
Select the TOP SURFACE of the 3D block.
Draw a circle of diameter 50mm.
Click Finish Sketch after completing.

STEP-6:
Select Extrude.
Select the sketch.
Click again to place the value.

Type in the value of 28mm.



Press ENTER to accept the value.

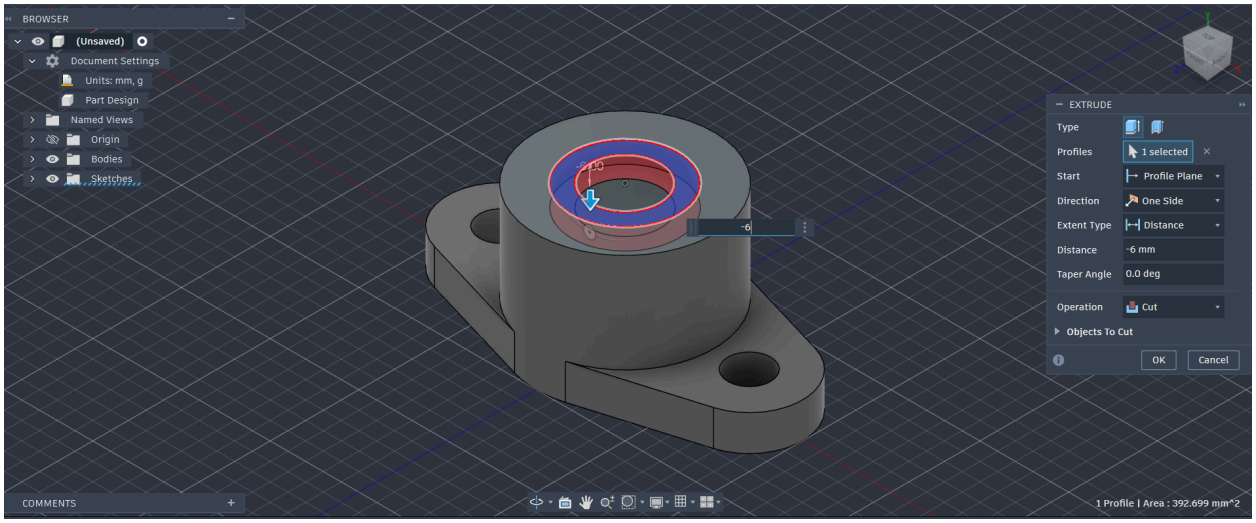
STEP-7:
Start the sketch command.
Select Sketch > Create Sketch.
Select the Sketch Plane.
Select the TOP SURFACE of the 3D block.
Draw a circle of diameter 20mm.
Click Finish Sketch after completing.

STEP-8:
Select Extrude.
Select the sketch.
Select operation cut.
Drag the arrow till the portion is cut out.

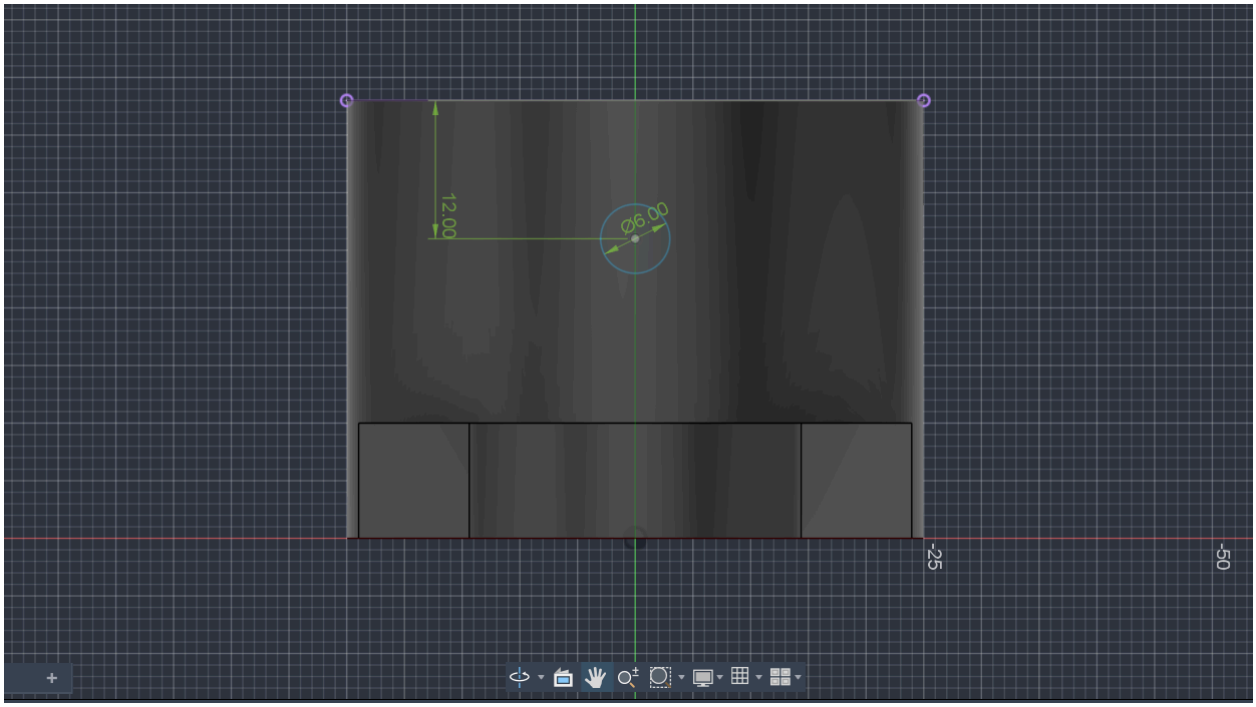
STEP-9:
Start the sketch command.
Select Sketch > Create Sketch.
Select the Sketch Plane.
Select the TOP SURFACE of the 3D block.
Draw a circle of diameter 30mm.
Click Finish Sketch after completing.

STEP-10:
Select Extrude.
Select the sketch.
Select operation cut.

Enter value of -6mm for that portion to be cut out.



STEP-11:
Start the sketch command.
Select Sketch > Create Sketch.
Select the Sketch Plane.
Select the left side of the 3D block.
Draw a circle of diameter 6mm.
Fix the distance of the circle from the top to be 12mm.



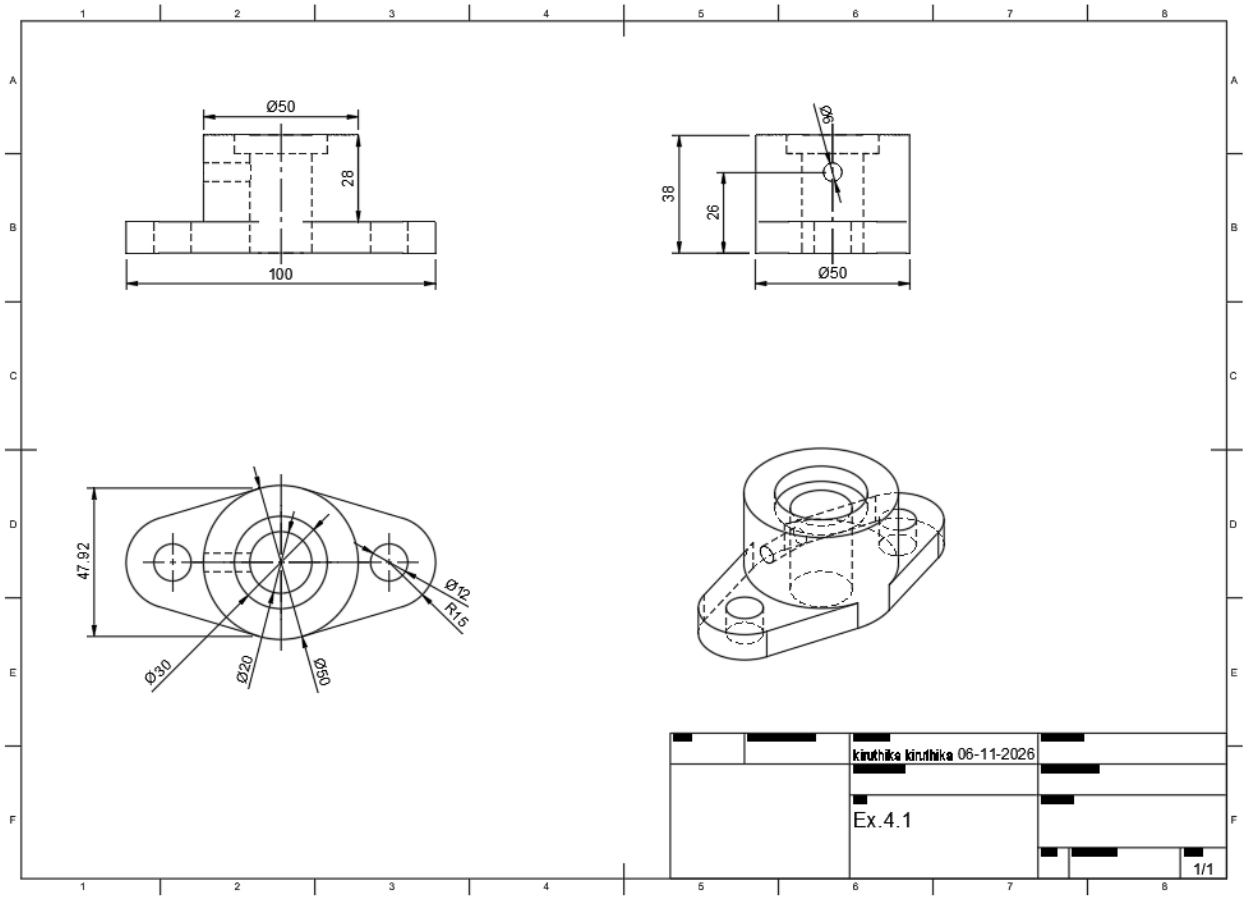
Click Finish Sketch after completing.

STEP-12:

Select Extrude.
Select the sketch.
Select operation cut.
Drag the arrow till the portion is cut out.

STEP-13:
Switch to drawing mode to plot the sketch in ISO standard.

OUTPUT:



RESULT:

Thus, the cut section for solid model is created.